Screenshots — Terraform Plugin Cache

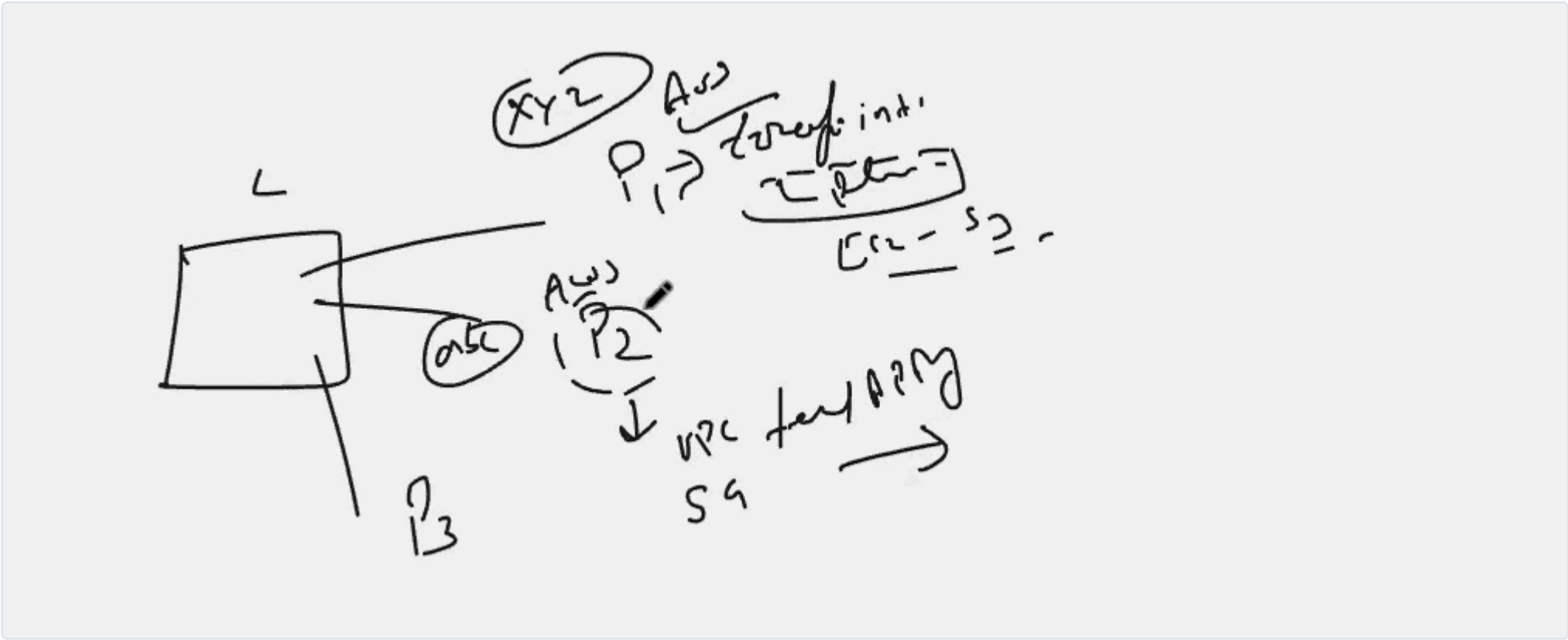
Lecture 5 · 17 shots, chronological · source stills from the session (2026-09-28 17:06 – 17:46)

01 / 17 Lecture definition — plugin cache is a local storage that saves plugins to make Terraform faster and more reliable `17:06:50`

- terraform plugin cache # Terraform plugin cache is like a **local storage that saves plugins** to make Terraform faster and more reliable.

The **Terraform plugin cache** is a built-in mechanism used **to store downloaded provider plugins in a centralized local directory so they can be reused across multiple Terraform projects**. OneUptime +1

By default, every time you run `terraform init`, Terraform downloads a completely separate copy of each required provider (such as the AWS, GCP, or Azure providers) directly into that specific project's `.terraform/providers` directory. Since provider binaries can easily exceed several hundred megabytes, downloading them repeatedly wastes significant time, network bandwidth, and disk space. OneUptime +1


XYZ
AWS
terraform init
P1
EC2 - S3
AWS
P2
VPC
SG
P3

```
root@cka-w2:~/project1# mkdir a b c
root@cka-w2:~/project1# ls
a  b  c
root@cka-w2:~/project1# cd a
root@cka-w2:~/project1/a# ls
root@cka-w2:~/project1/a# vim s3.tf
root@cka-w2:~/project1/a# ls
s3.tf
```

```
root@cka-w2:~/project1/a# cat s3.tf
resource "aws_s3_bucket" "first" {

  bucket = "linuxterraform12345"

  tags = {

    Name = "my_buck"

  }


}
```

```
root@cka-w2:~/project1/a# terraform plan

Error: Inconsistent dependency lock file

The following dependency selections recorded in the lock file are
inconsistent with the current configuration:
  - provider registry.terraform.io/hashicorp/aws: required by this configurat
ion but no version is selected

To make the initial dependency selections that will initialize the
dependency lock file, run:
  terraform init

root@cka-w2:~/project1/a#
```

Without a cache — one project's `.terraform` folder is 874M


```
[root@Devops-server a]# ls -a
.  ..  s3.tf  .terraform  .terraform.lock.hcl
[root@Devops-server a]# du -sh .terraform
874M    .terraform
```

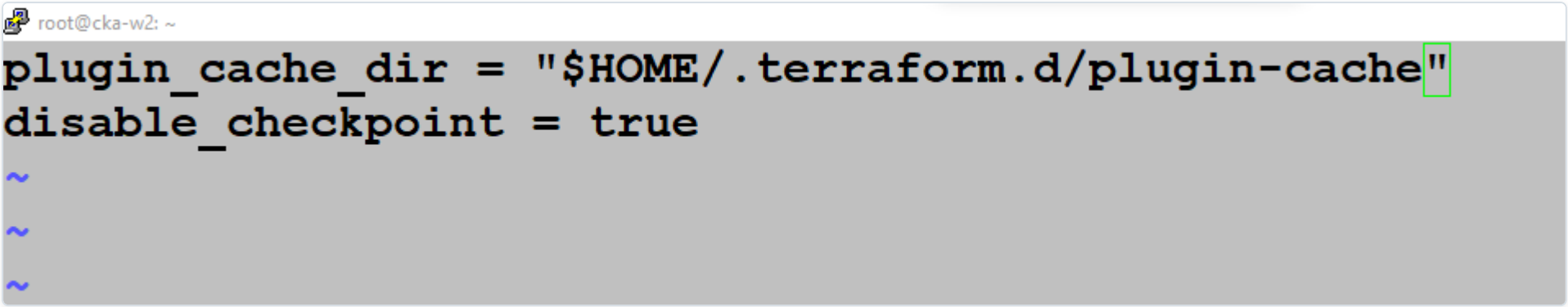
.terraformrc

Configuration File Syntax

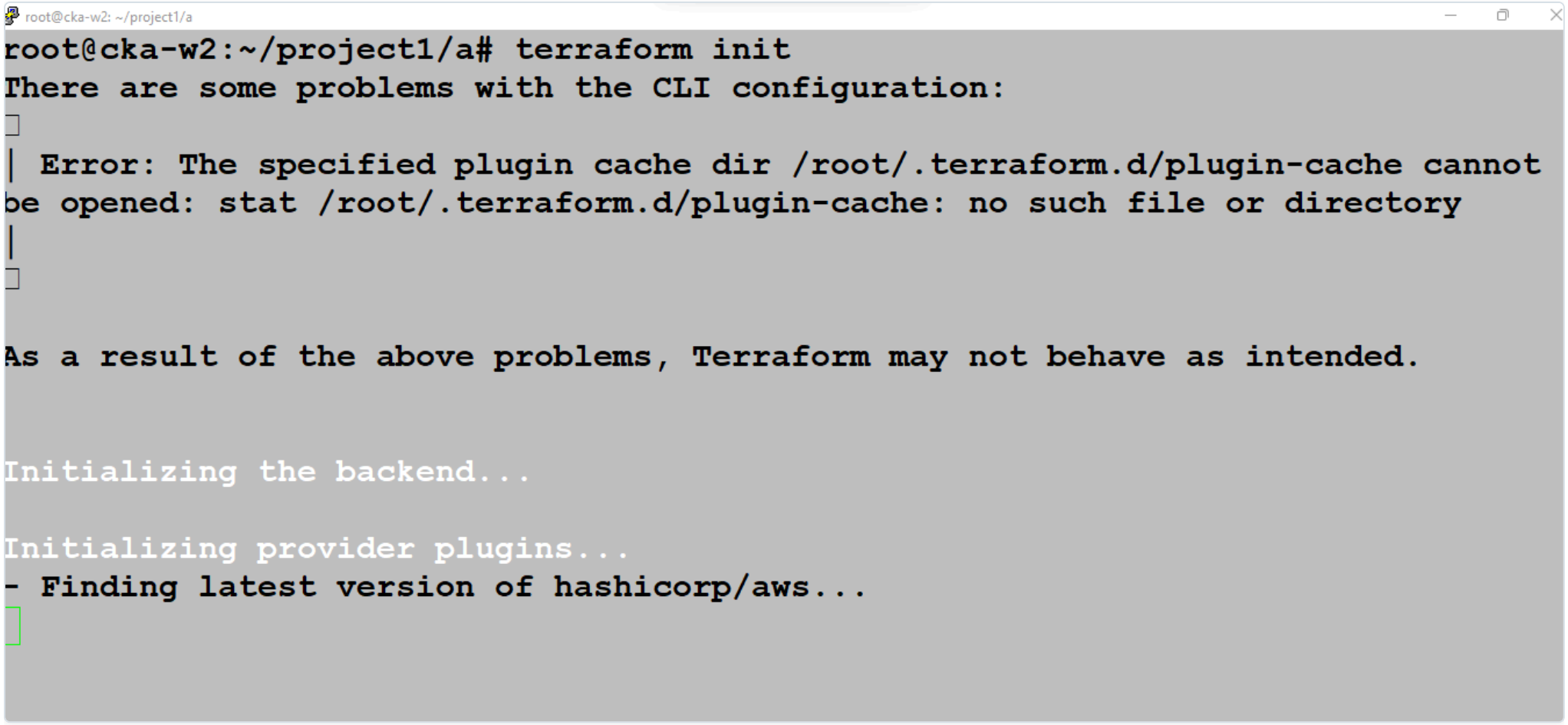
The configuration file uses the same *HCL* syntax as `.tf` files, but with different attributes and blocks. The following example illustrates the general syntax; see the following section for information on the meaning of each of these settings:

```
plugin_cache_dir   = "$HOME/.terraform.d/plugin-cache"
disable_checkpoint = true
```

~/.terraformrc contents in vim



```
root@cka-w2:~# touch .terraformrc
root@cka-w2:~# vim .terraform
root@cka-w2:~# vim .terraformrc
root@cka-w2:~# cd /root/.terraform.d
root@cka-w2:~/.terraform.d# ls
checkpoint_signature
root@cka-w2:~/.terraform.d# ls -ltrha
total 12K
-rw-r--r-- 1 root root  394 Sep 21 17:21 checkpoint_signature
drwxr-xr-x 2 root root 4.0K Sep 28 11:50 .
drwx------ 8 root root 4.0K Sep 28 12:01 ..
root@cka-w2:~/.terraform.d#
```


root@cka-w2: ~/project1/a
root@cka-w2:~/project1/a# terraform init
There are some problems with the CLI configuration:
Error: The specified plugin cache dir /root/.terraform.d/plugin-cache cannot be opened: stat /root/.terraform.d/plugin-cache: no such file or directory
As a result of the above problems, Terraform may not behave as intended.
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...

```
╷
│ Error: The specified plugin cache dir /root/.terraform.d/plugin-cache cannot
be opened: stat /root/.terraform.d/plugin-cache: no such file or directory
│
╵

As a result of the above problems, Terraform may not behave as intended.


Initializing the backend...

Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.66.0...
- Installed hashicorp/aws v6.66.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

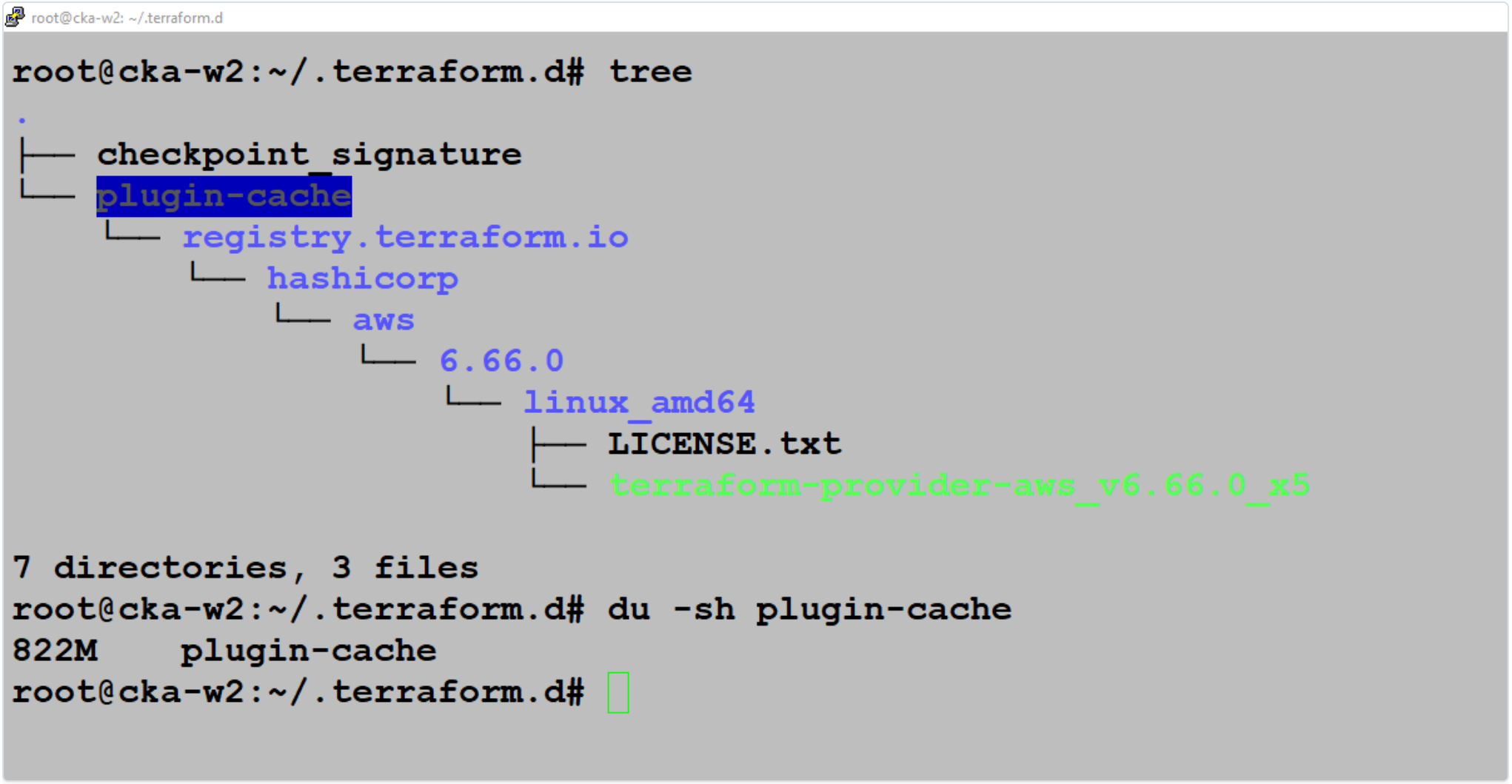
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
root@cka-w2:~/project1/a#
```

plugin-cache folder now present in ~/.terraform.d


```
root@cka-w2:~# cd .terraform.d
root@cka-w2:~/.terraform.d# ls
checkpoint_signature  plugin-cache
root@cka-w2:~/.terraform.d#
```

root@cka-w2: ~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp/aws/6.66.0/linux_amd64

```
root@cka-w2:~# cd .terraform.d
root@cka-w2:~/.terraform.d# ls
checkpoint_signature  plugin-cache
root@cka-w2:~/.terraform.d# cd plugin-cache/
root@cka-w2:~/.terraform.d/plugin-cache# ls
registry.terraform.io
root@cka-w2:~/.terraform.d/plugin-cache# cd registry.terraform.io/
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io# ls
hashicorp
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io# cd hashicorp/
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp# ls
aws
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp# cd aws/
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp/aws# ls
6.66.0
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp/aws# cd 6.66.0/
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp/aws/6.66.0# ls
linux_amd64
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp/aws/6.66.0# cd linux_amd64/
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp/aws/6.66.0/linux_amd64# ls
LICENSE.txt  terraform-provider-aws_v6.66.0_x5
root@cka-w2:~/.terraform.d/plugin-cache/registry.terraform.io/hashicorp/aws/6.66.0/linux_amd64#
```


root@cka-w2: ~/.terraform.d
root@cka-w2:~/.terraform.d# tree
.
├── checkpoint_signature
└── plugin-cache
    └── registry.terraform.io
        └── hashicorp
            └── aws
                └── 6.66.0
                    └── linux_amd64
                        ├── LICENSE.txt
                        └── terraform-provider-aws_v6.66.0_x5
7 directories, 3 files
root@cka-w2:~/.terraform.d# du -sh plugin-cache
822M   plugin-cache
root@cka-w2:~/.terraform.d#

Project `.terraform` — `linux_amd64` is a symlink into the cache, 0 files

